AI-Driven Mental Health Chatbot: Empowering Well-Being with Conversational AI and Retrieval-Augmented Generation

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***Abstract*—Mental health issues, including stress, anxiety, and depression, have seen a significant rise globally, impacting millions of lives and creating an urgent need for innovative solutions. While traditional mental health services are beneficial, barriers such as social stigma, high costs, and limited availability hinder widespread access. AI-driven chatbots offer a promising avenue for scalable, accessible mental health support. However, a notable research gap exists in developing chatbots that can generate empathetic, contextually accurate responses grounded in reliable mental health literature. In this study, two Retrieval-Augmented Generation (RAG) models are proposed: one employs all-MiniLM-L6-v2 embeddings with FAISS for efficient retrieval and OpenAI GPT for response generation, while the other leverages GoogleGenerativeAIEmbeddings, Pinecone for vector storage, and BART for response generation. Both models were implemented in Google Colab, designed to deliver personalized mental health assistance based on trusted resources. Experimental evaluation reveals that the chatbot produces highly empathetic and context-aware responses, effectively addressing user needs. The results show that our model achieved a 0.85 cosine similarity score.**

***Index Terms*—Anxiety, Depression, Generative Models, Retrieval-Augmented Generation (RAG), Vector Search, Google Generative AI Embeddings**

I. Introduction

Millions of individuals worldwide are impacted by mental health conditions like stress, anxiety, and depression each year, making them one of the main causes of disability. According to the World Health Organisation (WHO), depression affects more than 300 million people worldwide and is the primary cause of disability (WHO, 2021). Only a small percentage of people have access to quality mental health care, particularly in low- and middle-income nations, even if there are effective therapies available. The poor utilisation of current treatments is caused by a number of factors, including stigma, societal and cultural hurdles, and a lack of qualified mental health professionals. [1]

Artificial intelligence (AI), more especially Natural Language Processing (NLP) technologies, have become a viable answer to these problems. Chatbots and other AI-powered solutions have been created to offer scalable and easily accessible mental health assistance. [2], [3] These systems have a number of benefits, including as anonymity, accessibility, and the capacity to respond instantly, all of which can assist people in overcoming the stigma associated with asking for assistance [4]. With the use of evidence-based materials, chatbots can mimic therapeutic dialogues and provide assistance.

Traditional chatbots for mental health mostly used manual feature extraction techniques and basic response systems. These systems frequently produced generic or unnecessary answers because they were unable to grasp the complex context of user questions. Early systems, for instance, relied on static rule-based techniques or bag-of-words models that were unable to adequately comprehend user intent or dynamically adjust to discussions [4]. Although they offered rudimentary accessibility, they were less successful in offering significant emotional support because of their insufficient attention to contextual awareness.

Modern AI-based systems take advantage of developments in machine learning and natural language processing (NLP) to overcome these obstacles. Transformer-based chatbots, like BERT and GPT, have proven to be exceptionally adept at comprehending and producing contextually relevant responses. Transformer designs are excellent at recognising complex linguistic patterns because they are powered by attention mechanisms [5]. Their adaptability has further increased their relevance to a range of fields, including mental health support, as seen by unified text-to-text frameworks like T5 [6]. However, the use of many of these systems in delicate fields like mental health is limited because they are still limited

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to pre-trained models without incorporating domain-specific knowledge [7].

The Mental Health Chatbot presented in this paper is intended to help those who are looking for support for mental health conditions like depression and anxiety. [8] The chatbot combines the power of vector retrieval systems and huge language models through the use of Retrieval-Augmented Generation (RAG) techniques. The first model incorporates BART for response generation, Pinecone for vector storage, and GoogleGenerativeAIEmbeddings. The second model uses OpenAI GPT to generate replies, FAISS for effective retrieval, and all-MiniLM-L6-v2 embeddings. The chatbot offers sympathetic, scientifically supported answers by referencing reputable mental health materials like *Dr. David D. Burns' Feeling Good: The New Mood Therapy* and *Edmund J. Bourne's The Anxiety and Phobia Workbook* and *McDonagh, Barry - Dare The New Way to End Anxiety and Stop Panic Attacks*.

The structure of the paper is as follows: Section II provides an overview of the relevant research on chatbots for mental health. Section III explains the methodology used to develop the chatbot. In Section IV, cosine similarity and perplexity are introduced as metrics to evaluate the performance of the models and the precision of the generated responses. Section V concludes the paper, summarizing the main findings and future directions.

II. Related work

Anxiety, sadness, and other mental health conditions are worldwide health concerns that impact millions of individuals annually. Digital health solutions have become a viable means of reducing some of these obstacles in recent years. Artificial intelligence (AI)–powered chatbots have gained popularity as a possible way to assist mental wellness. Chatbots can lessen the stigma attached to traditional mental health services by providing consumers with a discreet, easily accessible, and frequently anonymous means of requesting assistance. These systems imitate human-like discussions using machine learning (ML) and natural language processing (NLP), offering emotional support and helping people develop coping mechanisms for mental health issues [4].

The development of chatbots has advanced even more with the introduction of transformer-based models like BERT (Bidirectional Encoder Representations from Transformers) and GPT (Generative Pretrained Transformers) [9]. These models, which are highly effective at comprehending and producing language, are especially well-suited for use in mental health applications because of their capacity to comprehend context, absorb complex human emotions, and respond with empathy [7], [10]. Retrieval-augmented generation (RAG) model integration has made it possible for chatbot systems to communicate with users in a more precise, tailored, and context-aware manner in collaboration with AI accomplishments.

Through conversational therapy, symptom tracking, and emotional support, chatbots have been used to help people with stress, anxiety, and depression. Early attempts concentrated on rule-based systems, such ELIZA [11], which employed pattern matching to replicate simple therapeutic interactions. Nevertheless, the capabilities of these systems were restricted, and they frequently did not yield significant results.

Deep learning and Natural Language Processing (NLP) have greatly increased the efficacy of chatbots for mental wellness. The creation of chatbots driven by transformer-based models, such BERT and GPT, is one of the most notable developments in this area. These chatbots have transformed natural language processing (NLP) tasks including sentiment analysis, text categorisation, and language production. These models have relevance for mental health applications because they provide better language creation and understanding skills, enabling more complex, sympathetic, and contextually relevant responses.

The effectiveness of AI-powered chatbots in mental health has been shown in numerous studies. For instance, Woebot, a chatbot designed to help people with anxiety and depression, has demonstrated potential in offering successful psychological therapies and is based on the ideas of Cognitive Behavioural Therapy (CBT) [12].

Another well-known chatbot for mental health, Wysa, uses AI and natural language processing to provide self-help resources and emotional support. By giving users a private, anonymous forum to talk about their mental health issues.

Woebot and Wysa help to lessen the stigma attached to getting treatment.

The performance of chatbots has been significantly improved by the incorporation of retrieval-augmented models. A study by [13] presented a retrieval-based methodology for producing context-aware answers by obtaining pertinent data from extensive knowledge bases, such internet forums or medical literature. To produce more precise and customised results, this model was coupled with a generative model. This approach of Retrieval-Augmented Generation (RAG) has been employed in several mental health chatbots, enabling them to provide responses grounded in evidence-based therapeutic strategies.

The use of vector databases for effective information retrieval is another significant advancement in chatbot technology for mental health. In order to guarantee that chatbots deliver accurate and relevant information, two well-known systems—Pinecone and FAISS—enable quick, scalable retrieval of semantically related documents from big databases [14], [15]. These methods enable chatbots to better comprehend customer enquiries and get the most relevant mental health resources by integrating vector search with embeddings. Studies show that by offering basic support and direction to those in need, chatbots for mental health assistance might lessen the workload for mental health experts. According to [12], chatbots can serve as a bridge before people seek professional assistance by providing first support for those with mild to moderate symptoms of anxiety and depression. Additionally, chatbots offer a widely available, reasonably priced alternative for people who might not otherwise be able to get mental health care because of financial or geographic constraints.

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TABLE I
COMPARISON OF NOTABLE MENTAL HEALTH CHATBOTS

Chatbot	Model Type	Therapeutic Basis	Key Features	Key Findings / Limitations
Woebot	ML-driven AI (GPT, BERT)	Cognitive Behavioral Therapy (CBT)	Mood tracking, CBT-based interventions, evidence-based self-help resources	Effective for anxiety and depression [12]; Limited for crisis situations; lacks deep flexibility in responses.
Wysa	ML-driven AI (BERT)	Emotion-focused Therapy	Emotional support, self-help resources, personalized conversations	Provides emotional support [16]; Limited in handling complex mental health issues.
ELIZA	Pattern-matching Algorithm	None	Simulated psychotherapy conversations based on pre-defined rules	Limited interaction capabilities [11]; Superficial conversations; unable to understand or process complex emotional responses; lacks advanced therapeutic capabilities.

Despite their promising potential, chatbots for mental health have a number of significant drawbacks. First, the scope of help is still restricted; Woebot and other platforms expressly warn users that they are not prepared to deal with emergency circumstances and cannot take the place of professional therapy. Furthermore, rule-based systems such as Wysa and Woebot have limited response flexibility because they offer ”safe” preapproved responses, which can occasionally result in useless advise when user input is misread. Users frequently voice ethical concerns about data security and privacy, as well as the possible negative consequences of AI-driven treatments in delicate fields like mental health. Additionally, even while chatbots can help people express their emotions, it is still unclear if they can promote true healing and offer ongoing assistance, particularly when contrasted with real therapists. These restrictions highlight the need for ongoing research to enhance the effectiveness, ethical integrity, and trustworthiness of AI-powered mental health solutions.

Comparison of Related Works

A comparison of existing chatbots, focusing on key attributes is presented in Table I. This comparison highlights the strengths and limitations of each system.

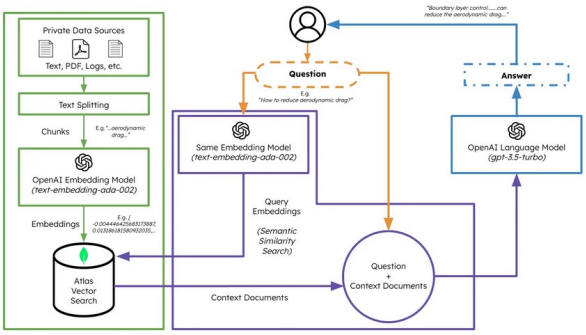


Fig. 1. Architecture diagram

III. METHODOLOGY

This section outlines the steps taken in the creation and assessment of the mental health chatbot, which is intended to provide assistance to those dealing with stress, anxiety, and other mental health conditions. To give a comprehensive understanding of the chatbot's operation and design, Fig. 2 displays the system architecture. It consists of the fundamental elements that function in unison, including response production, retrieval strategies, embedding construction, and data preprocessing was performed. Three crucial phases are the emphasis of the methodology: Data preparation, system evaluation, and model design. To guarantee that the chatbot can deliver accurate, sympathetic, and personalized responses, each of these steps is essential.

A. Data Prepration

1) Data Collection: Data is essential to creating a successful chatbot, especially one that offers mental health assistance. When it comes to mental health, the quality and applicability of the data have a direct impact on the chatbot's comprehension and contextual, accurate, and sympathetic responses to user concerns. Talking about mental health is extremely delicate and calls for a thorough comprehension of psychological ideas, sympathetic reactions, and the capacity to handle a range of emotional states. Therefore, to guarantee that the chatbot can provide knowledgeable and beneficial assistance, the data used must be both authoritative and diversified.

For this project, data was sourced from three authoritative texts in the field of mental health are: [17], [18], [19]
These books were chosen because they offer therapeutic, evidence-based methods for treating depression and anxiety.

B. Data Preprocessing

One of the most important steps in getting the raw data ready for chatbot training is data preprocessing. To make sure it was appropriate for entering into the model, the data taken from the books went through a number of preparation steps in this research.

1) Data Extraction: Python modules such as pymupdf and PyPDF2 were used to extract the data from the books. This made it possible to extract textual content from the books' PDF versions. Because of their effectiveness in managing structured documents, such as books, maintaining text formatting, and guaranteeing precise extraction, these technologies were chosen. The content was transformed into plain text (.txt) files for additional processing after the text was extracted.

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2) Data Cleaning: To make sure the input data is reliable, consistent, and prepared for model training, data cleaning is an essential step. In order to increase the quality of the data supplied into the chatbot and eventually boost its performance, this step focuss on eliminating unnecessary aspects from the raw input. Page numbers, chapter headers, references, footnotes, and other unnecessary details were frequently included in the excerpted material from the books. To concentrate only on the information pertinent to mental health support, these elements were eliminated using regular expressions. The text occasionally contained formatting flaws, encoding problems, and special characters throughout the extraction procedure. To guarantee consistency across the text, for instance, non-ASCII characters and line breaks were eliminated. The text is converted into lowercase for standardization. Using libraries like NLTK and spaCy, common words like "the," "and," "is," etc.—referred to as stopwords—were eliminated from the data. By lowering data noise and enabling the model to concentrate on the more significant words in phrases, stopword elimination enhances the chatbot's capacity to produce accurate responses.

C. Chunking

The RecursiveCharacterTextSplitter approach is used to chunk large textual content in order to efficiently manage and analyse it. This method preserves important context while breaking the material up into smaller, more manageable sections. As illustrated in Fig. 1, It ensures that each chunk stays within a predetermined size by recursively dividing the text according to character length. Large papers like Feeling Good: The New Mood Therapy and The Anxiety and Phobia Workbook benefit greatly from this since it makes it possible to consume them in digestible chunks without losing crucial context.

By ensuring that the material is divided into meaningful sections, this technique raises the effectiveness of subsequent procedures like model training and embedding. Because it can handle big documents while preserving the integrity of sentences and paragraphs, which is essential for the chatbot to generate coherent and contextually relevant responses.

D. Embedding and Retrieval

The RecursiveCharacterTextSplitter was used to chunk the text, and then the text was transformed into numerical representations that the model could comprehend.As illustrated in Fig. 3, The text is converted into vectorised forms by this process, called embedding, which makes them better suited for machine learning models [20], [21]. For effective retrieval and similarity search, I used two distinct embedding techniques, each paired with a corresponding vector store.

Embedding Equation:

$$v = \text{Embedding}(x) \tag{1}$$

Where:

x is the input text

v is the resulting vector in the continuous vector space.

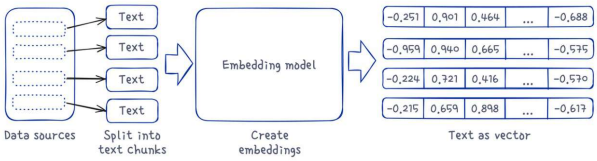


Fig. 2. Illustration of Chunking and Embedding procedure

1) Model 1: Pinecone and GoogleGenerativeAIEmbeddings: The GoogleGenerativeAIEmbeddings embeddings record detailed semantic links between words and sentences and are tuned for natural language comprehension. Pinecone, a high-performance vector database that is especially well-suited for extensive similarity searches, is where the embeddings were kept. Even with big datasets, Pinecone's highly scalable architecture makes it possible to get embeddings rapidly and effectively, guaranteeing that the chatbot can provide appropriate replies in a timely manner.

With this combination, Model 1 can handle a variety of user inputs and respond according to semantic similarity, preserving context in discussions about mental health support.

2) Model 2: FAISS-enabled all-MiniLM-L6-v2 embeddings: I used all-MiniLM-L6-v2 embeddings for Model 2, a lightweight yet effective sentence-level embedding model. The meaning of sentences and their interactions are well captured by these embeddings. We used FAISS (Facebook AI Similarity Search), a very effective similarity search library that facilitates quick retrieval of the most similar embeddings, to handle the vast number of embeddings. Model 2 can promptly deliver the most contextually relevant answers to user questions thanks to FAISS's optimisation for high-performance searches in big databases.

Model 2 combines FAISS with the all-MiniLM-L6-v2 embeddings to provide high-quality, context-aware replies with computational efficiency.

E. Model Design and Response Generation

Response generation is a critical pipeline phase that comes after embedding and retrieval. In this step, pre-trained generative models are used to generate contextually accurate and meaningful replies depending on the chunks that were obtained. To assess how well they produced replies for mental health support, two models with different configurations were created:

1) Model 1: Pinecone and GoogleGenerativeAIEmbeddings in BART-Large: In this model GoogleGenerativeAIEmbeddings were utilised for chunk embeddings, which were then stored and effectively retrieved with Pinecone.

Generation: Facebook/bart-large, a transformer-based generative model, received the extracted text segments.

Why BART? Text creation and other sequence-to-sequence activities, such as summarisation, benefit greatly from the use of BART (Bidirectional and Auto-Regressive Transformers).

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Because of its architecture, it can rebuild and produce cohesive responses, guaranteeing that user enquiries are addressed in a clear and pertinent manner.

2) Model 2: OpenAI GPT with FAISS and all-MiniLM-L6-v2: In this model all-MiniLM-L6-v2 is used for embedding generation and FAISS used to store and retrieve these embeddings.

Generation: OpenAI GPT, a state-of-the-art generative model, was fed the extracted text passages.

Why GPT? Generative Pre-trained Transformer, or GPT, is very good at creating text that looks human. It is especially well-suited for conversational agents in delicate domains like mental health assistance because of its extensive training on a variety of data, which allows it to comprehend spoken language efficiently and produce contextually relevant responses.

In order to concentrate on the most relevant portions of the input sequence during both encoding and decoding, these models mainly rely on the attention mechanism.

Formally, the scaled dot-product attention function can be used to represent the attention mechanism:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \qquad (2)$$

Where The query vector Q, represents the current word or token, the key vector K, represents the context or words that came before it, the value vector V, represents the material that has to be addressed, the dimension of the key vectors is dk.

This formula shows how the model calculates a weighted sum of the value vectors VV, where the query and key vectors' similarity determines the weights. By ensuring that these weights add up to one, the softmax function enables the model to focus on various input sequence segments with varying degrees of significance. This method is essential to the model's ability to produce responses that are both logical and related to the context.

F. Pipeline Integration

Fig. 4 shows the Retrieval-Augmented Generation (RAG) pipeline is used by both models. Text is integrated into vector representations after being chunked. Similarity search is used to retrieve relevant chunks (Pinecone/FAISS). The final answer is generated by feeding the retrieved portions into the corresponding generative models (BART or GPT).

G. Justification for the Dual-Model Design

The purpose of the two models was to evaluate the performance of different embedding techniques (GoogleGenerativeAIEmbeddings vs. all-MiniLM-L6-v2) and generative models (BART vs. GPT).

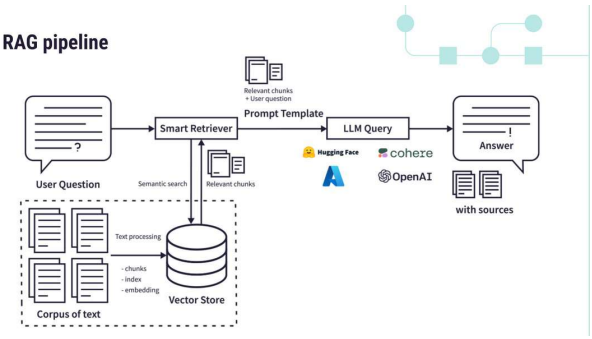


Fig. 3. RAG pipeline visualization

By integrating excellent embeddings, efficient retrieval methods, and complex generative models—all crucial for responding to enquiries about mental health—both systems aim to deliver trustworthy and context-aware responses. Utilising each model's advantages, the system can provide customised answers that are more in line with the user's mental health needs, ensuring that individuals receive relevant and empathetic support.

IV. Models Evaluation using Cosine Similarity and Perplexity

I used Cosine Similarity and Perplexity to assess the performance of the two models: Pinecone and GPT with all-MiniLM-L6-v2 embeddings and FAISS, and BART with GoogleGenerativeAIEmbeddings. The degree of similarity between the generated responses and the user query or a reference response is measured by cosine similarity. A higher score means that the model's response is more semantically and contextually aligned with the input. The score ranges from 0 to 1. Another popular method to evaluate performance is Perplexity calculation. In language modelling, Perplexity is frequently used to assess how effectively a probabilistic model forecasts a sample. It is commonly used in natural language processing (NLP) activities such as text generation, and it can be especially helpful when evaluating the level of responses produced by various models.

A. Evaluation using Cosines similarity

The first step in evaluating the models is to create embeddings for the input query and the models' associated answers. To produce the appropriate outputs, the user query is first run through the models:

Model 1: response generated by BART

Model 2: reaction provided by GPT

Following the generation of the replies, pre-trained embedding models are used to convert these outputs into numerical vector representations. In particular, the responses from Model 1 are generated using GoogleGenerativeAIEmbeddings, and the responses from Model 2 are generated using all-MiniLM-L6-v2. Both the user query and the model-generated responses' semantic content are represented in vector space

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by these embeddings. By comparing their embeddings, we can then determine how comparable the user query and the model's output are, enabling us to evaluate the relevance and alignment of the responses with the input query.

To calculate the Cosine Similarity the following formula is used:

$$\text{Cosine Similarity} = \frac{\|A\|\|B\|}{A \cdot B} \tag{3}$$

Here:

A: The user query in vectorised form

B: The model-generated answer in vectorised form

1) Results: Cosine similarity was computed between the responses produced by both models and the user inquiry in order to assess the chatbot's performance. This measure assists in measuring how closely the generated answers match the query's context. As shown in Table II, Model 2 (OpenAI GPT with FAISS and all-MiniLM-L6-v2) [22] has generated response more similar to the query as compared to the Model 1 (Pinecone and GoogleGenerativeAIEmbeddings in BART-Large) [23].

TABLE II
COSINE SIMILARITY RESULTS FOR MODEL EVALUATION

Cosine Similarity Results	
Model Name	Similarity Score
Model 1	0.7217
Model 2	0.8521

B. Evaluation using Perplexity

The perplexity metric measures how effectively a model predicts a word sequence in order to evaluate the quality of text production. First, the original query and the response produced by the model are fed into the evaluation system. Based on the probability distribution it gives the word sequence, perplexity gauges how confident the model is in producing the specified output.

The model determines the likelihood that the sequence will be formed for every response. Reduced perplexity scores show that the generated language is fluid and contextually appropriate, and that the model is more confident in its predictions.

To calculate the Perplexity first we calculate the logarithmic likelihood of the response. The formula used for this is as follows:

$$H(p) = -\frac{1}{N}\sum_{i=1}^{N}\log(p(w_i)) \tag{4}$$

Where wi is the i-th word in the response, and p(wi) is the probability of the i-th word.

After calculating likelihood, we calculate Perplexity using the following formula:

$$\text{Perplexity} = 2^{H(p)} \tag{5}$$

1) Results: To assess the chatbot ability to produce text that is both contextually relevant and coherent, perplexity was calculated for the responses generated by both models. According to Table III, Model 2 (OpenAI GPT with FAISS and all-MiniLM-L6-v2) generates responses that are linguistically fluent and in line with natural language patterns, while Model 1 (Pinecone and GoogleGenerativeAIEmbeddings in BART-Large) shows a lower perplexity score. Lower perplexity values indicate higher accuracy.

TABLE III
PERPLEXITY CALCULATION RESULTS FOR MODEL EVALUATION

Perplexity calculation Results	
Model Name	Model Perplexity
Model 1	27.139
Model 2	53.362

A graphical representation of the evaluation criteria is included in Figure 5 to give a more transparent comparison of the two models' performances. The graph illustrates how the two models differ in terms of fluency (Perplexity) and semantic similarity (Cosine Similarity).

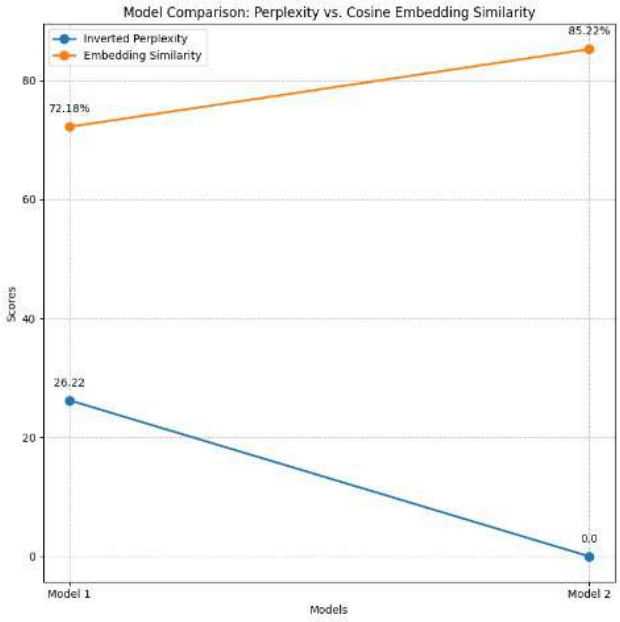


Fig. 4. Comparative visualization of Cosine Similarity and Perplexity scores

Comparison with Existing Models: Table IV presents a comparison between our best model and other existing approaches in terms of Cosine Similarity and Perplexity. These metrics allow us to evaluate the relative performance and effectiveness of our model in providing accurate and contextually relevant responses.

V. CONCLUSION

In order to give people appropriate information and assistance for managing mental health issues, specifically stress,

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TABLE IV
COMPARISON OF MODEL 2 WITH OTHER WORKS IN TERMS OF COSINE SIMILARITY AND PERPLEXITY

Model Name	Cosine Similarity	Perplexity
Model 2 (FAISS-enabled all-MiniLM-L6-v2)	0.8521	53.362
Denecke et al. (2021) AI Chatbot	0.6500	78.455
Boucher et al. (2021) AI-Powered Chatbot	0.5000	95.372
Adamopoulou et al. (2020) Overview Model	0.6300	82.341

anxiety, and depression, this article described the design, development, and evaluation of a mental health chatbot. To make sure it offers meaningful and contextually relevant interactions, the chatbot employs a retrieval-augmented generation (RAG) technique that combines embedding-based retrieval and response generation processes [24]. Two models were compared: Model 2, which used OpenAI GPT, and Model 1, which relied on BART for response creation. Model 2 performed better than Model 1 in terms of response quality and relevance, according to a cosine similarity evaluation, proving the usefulness of contemporary language models in applications for mental health care.

Subsequent research will try to improve these models by adding more contextual knowledge and fine-tuning datasets unique to mental health. The relevancy of the chatbot will be further increased by improving its personalization capabilities to modify responses according to user preferences and past interactions. By putting in place a framework for continual learning, the chatbot might develop and offer increasingly useful assistance over time. Last but not least, resolving ethical issues such as data privacy and coordinating the chatbot's responses with professional advice on mental health would be essential for its practical implementation.

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